Decision Analysis Theory

Decision Making with Multiple Objectives 2

Mathematical Programming Methods

- Goal Programming.
  - Problems which do not include a specific objective function, but rather a collection of goals to be achieved.
- Multiple Objective Optimization.
  - Problems which include more than one objective function.

- Both methods require an iterative solution procedure in which the decision maker investigates a variety of solutions to find one that is most satisfactory.
  - Might need to solve several variations of the problem before arriving to an acceptable solution.

Mathematical Programming Methods

- Linear Programming.

$$
\begin{aligned}
&\text{MAX (or MIN)} && f_0(X_1, X_2, \ldots, X_n) \\
&\text{Subject to} && f_1(X_1, X_2, \ldots, X_n) \le b_1 \\
& && f_2(X_1, X_2, \ldots, X_n) \ge b_2 \\
& && \ldots \\
& && f_m(X_1, X_2, \ldots, X_n) = b_m
\end{aligned}
$$

$$
\begin{aligned}
&\text{MAX (or MIN)} && c_1X_1 + c_2X_2 + \ldots + c_nX_n \\
&\text{Subject to} && a_{11}X_1 + a_{12}X_2 + \ldots + a_{1n}X_n \le b_1 \\
& && a_{21}X_1 + a_{22}X_2 + \ldots + a_{2n}X_n \ge b_2 \\
& && \ldots \\
& && a_{m1}X_1 + a_{m2}X_2 + \ldots + a_{mn}X_n = b_m
\end{aligned}
$$

Mathematical Programming Methods

- Integer Programming.
  - Additional constraints.

$$X_1, X_2, \ldots, X_n \text{ integers.}$$

- Nonlinear Programming
- Mixed-Integer Programming
- Quadratic Programming
- Dynamic Programming

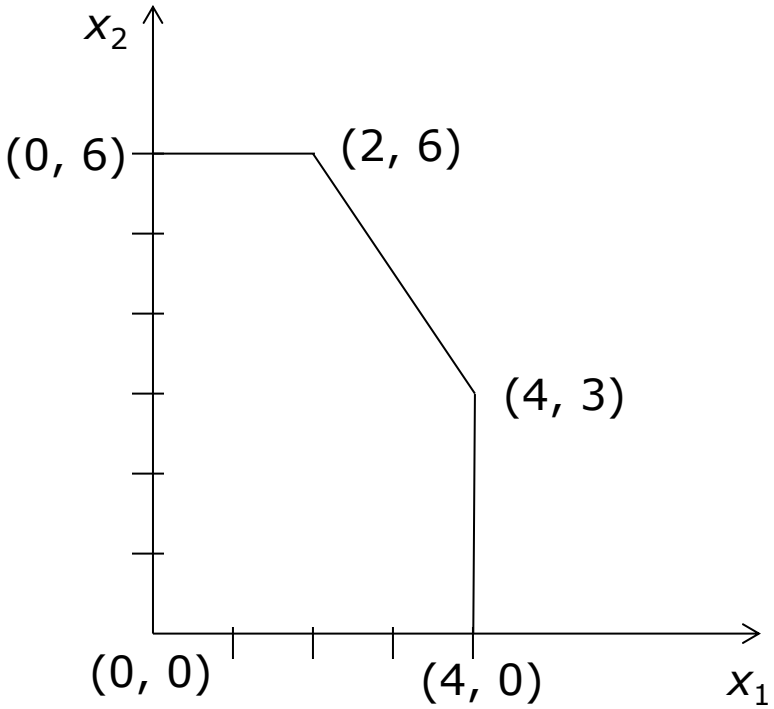
Linear Programming

- Three key components in a LP model.
  - Objective function (has to be linear: minimize, maximize).
  - Decision variables.
  - Constraints (has to be linear combination).

Max $Z = 3x_1 + 5x_2$

Subject to

$$x_1 \le 4$$
$$2x_2 \le 12$$
$$3x_1 + 2x_2 \le 18$$
$$x_1 \ge 0,\ x_2 \ge 0$$



Linear Programming Example 1

- An investor has money-making activities *A* and *B* available at the beginning of each of the next 5 years, while activities *C* and *D* are available only once.
  - Each $1 invested in *A* at the beginning of one year returns $1.40 two years later.
  - Each $1 invested in *B* at the beginning of one year returns $1.70 three years later.
  - Each $1 invested in *C* at the beginning of second year returns $2.00 four years later.
  - Each $1 invested in *D* at the beginning of fifth year returns $1.30 one year later.
- The investor begins with $10,000 and wishes to know which investments to make so as to maximize accumulated wealth at the beginning of the sixth year.

Linear Programming Example 1

- Decision variables:
  - A_t = amount invested in A at the beginning of year t (t = 1, 2, 3, 4)
  - B_t = amount invested in B at the beginning of year t (t = 1, 2, 3)
  - C_2 = amount invested in C at the beginning of year 2.
  - D_5 = amount invested in D at the beginning of year 5.
  - R_t = amount held in reserve from year t until year t+1 (t = 1, 2, 3, 4).

Amount invested	Amount available
$A_1 + B_1 + R_1$	10,000
$A_2 + B_2 + C_2 + R_2$	R_1
$A_3 + B_3 + R_3$	$R_2 + 1.4A_1$
$A_4 + R_4$	$R_3 + 1.4A_2 + 1.7B_1$
$D_5 + R_5$	$R_4 + 1.4A_3 + 1.7B_2$
	$R_5 + 1.4A_4 + 1.7B_3 + 2.0C_2 + 1.3D_5$

Linear Programming Example 1

- Objective function and constraints:

Max $R_5 + 1.4A_4 + 1.7B_3 + 2.0C_2 + 1.3D_5$

Subject to

$$A_1 + B_1 + R_1 = 10{,}000$$
$$A_2 + B_2 + C_2 + R_2 = R_1$$
$$A_3 + B_3 + R_3 = R_2 + 1.4A_1$$
$$A_4 + R_4 = R_3 + 1.4A_2 + 1.7B_1$$
$$D_5 + R_5 = R_4 + 1.4A_3 + 1.7B_2$$
$$A_t, B_t, C_2, D_5, R_t \geq 0$$

Linear Programming Example

- Airline crew scheduling problem.
  - An airline company must decide how many new flight attendants to hire and train over the next six months.
  - Required number of hours to be covered for the airline:

Month	**1**	**2**	**3**	**4**	**5**	**6**
Requirement	8000	9000	7000	10000	9000	11000

  - Objective: find a plan for hiring which minimizes total costs.

Linear Programming Example

- Airline crew scheduling problem.
  - Other requirements:
    - New hires need 1 month of training.
    - Trainees need 100 hours of supervision by experienced flight attendants.
    - Experienced flight attendants work 150 hours per month.
    - Need 60 experienced flight attendants at the beginning of first month.
    - 10% of experienced flight attendants quit each month.
    - Salaries (costs) include $1800/month for trainees, and $3400/month for experienced flight attendants.

Linear Programming Example

- Airline scheduling problem.
  - Y_t = number of flight attendants hired at the beginning of month t, t = 1, 2, …, 6
  - X_t = number of experienced flight attendants available for flight in month t, t = 1, 2, …, 6
  - Objective function:

$$\text{MIN } 1800Y_1 + 3400X_1 + 1800Y_2 + 3400X_2 + 1800Y_3 + 3400X_3 + 1800Y_4 + 3400X_4 + 1800Y_5 + 3400X_5 + 1800Y_6 + 3400X_6$$

  - Subject to:

$$Y_t \geq 0$$

$$X_t \leq 0$$

Linear Programming Example

- Airline scheduling problem.
  - Subject to:

$$100Y_1 + 150X_1 \geq 8000$$
$$100Y_2 + 150X_2 \geq 9000$$
$$100Y_3 + 150X_3 \geq 7000$$
$$100Y_4 + 150X_4 \geq 10000$$
$$100Y_5 + 150X_5 \geq 9000$$
$$100Y_6 + 150X_6 \geq 11000$$
$$X_1 = 60$$
$$Y_1 \leq 60$$
$$Y_2 \leq Y_1 + 0.9\ X_1$$
$$Y_3 \leq Y_2 + 0.9\ X_2$$
$$Y_4 \leq Y_3 + 0.9\ X_3$$
$$Y_5 \leq Y_4 + 0.9\ X_4$$
$$Y_6 \leq Y_5 + 0.9\ X_5$$

Dynamic Programming

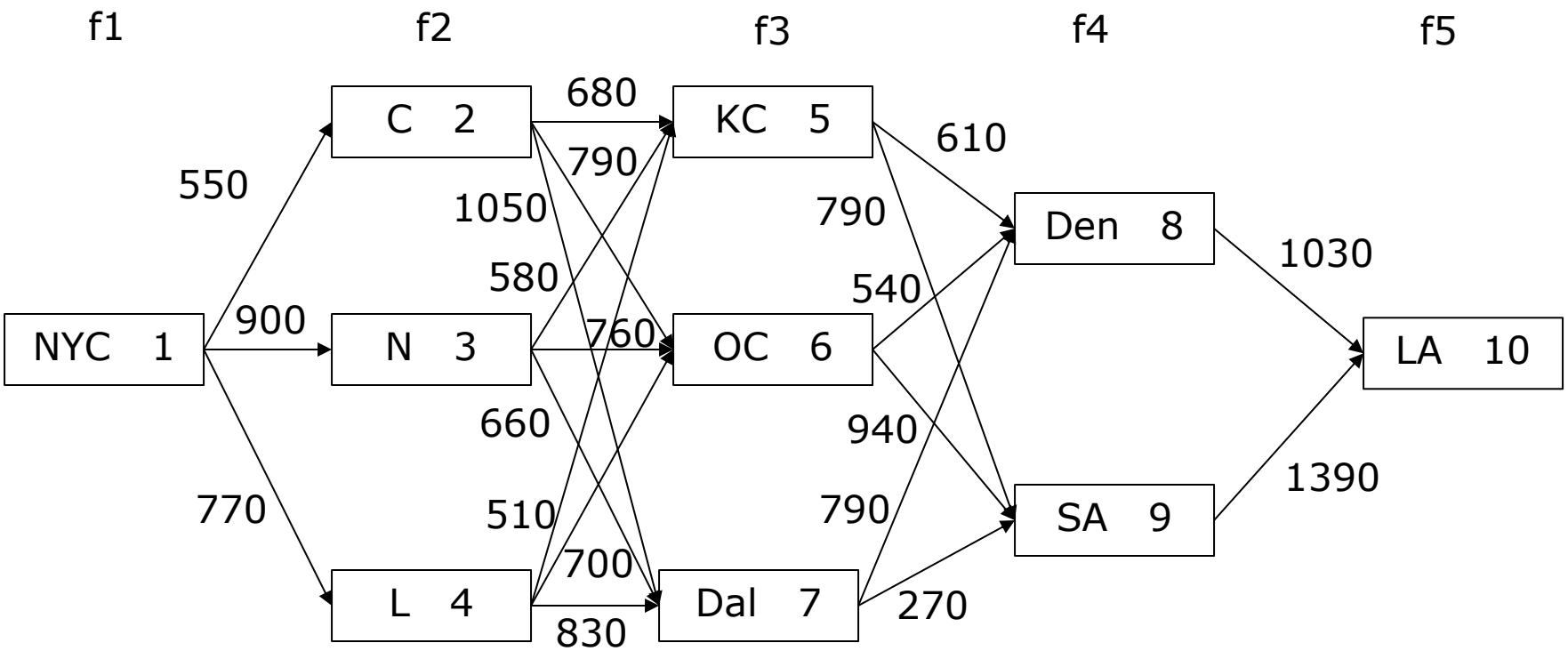
- Mathematical technique used for making a sequence of interrelated decisions.
  - Systematic procedure to determine the optimal combination of decisions.
  - No standard mathematical formulation.

Mathematical Programming Methods

- Dynamic programming.
  - Optimization technique which provides optimal solutions.
  - Make a sequence of interrelated decisions.
  - No standard formulation as in LP.
  - Developed to fit each case study.
  - Divide problem into stages with a policy decision.
  - Each stage has a number of states.
  - Find optimal policy at each stage for each of the possible states.
  - Given the current state, an optimal policy for the remaining stages is independent of the policy adopted in previous stages.
  - Solution approach begins by finding the optimal policy for the last stage.

Dynamic Programming

- How to drive from NYC to LA on the shortest path?



Dynamic Programming

- How to drive from NYC to LA on the shortest path?
- f4 to f5:
  - **<u>f4(8) = c810 = 1030</u>**
  - f4(9) = c910 = 1390
- f3 to f4:
  - **<u>f3(5) = c58 + f4(8) = 310 + 1030 = 1640</u>**
  - f3(5) = c59 + f4(9) = 790 + 1390 = 2180

  - **<u>f3(6) = c68 + f4(8) = 540 + 1030 = 1570</u>**
  - f3(6) = c69 + f4(9) = 940 + 1390 = 2330

  - f3(7) = c78 + f4(8) = 790 + 1030 =1820
  - **<u>f3(7) = c79 + f4(9) = 270 + 1390 =1660</u>**

Dynamic Programming

- How to drive from NYC to LA on the shortest path?
- f2 to f3:
  - **<u>f2(2) = c25 + f3(5) = 680 + 1640 = 2320</u>**
  - f2(2) = c26 + f3(6) = 790 + 1570 = 2360
  - f2(2) = c27 + f3(7) = 1050 + 1660 = 2710

  - **<u>f2(3) = c35 + f3(5) = 580 + 1640 = 2220</u>**
  - f2(3) = c36 + f3(6) = 760 + 1570 = 2330
  - f2(3) = c37 + f3(7) = 660 + 1660 = 2320

  - **<u>f2(4) = c45 + f3(5) = 510 + 1640 = 2150</u>**
  - f2(4) = c46 + f3(6) = 700 + 1570 = 2270
  - f2(4) = c47 + f3(7) = 830 + 1660 = 2490

Dynamic Programming

- How to drive from NYC to LA on the shortest path?
- f1 to f2:
  - **<u>f1(1) = c12 + f2(2) = 550 + 2320 = 2870</u>**
  - f1(1) = c13 + f2(3) = 900 + 2220 = 3120
  - f1(1) = c14 + f2(4) = 770 + 2150 = 2920

  - Shortest path: NYC – C – KC – Den – LA: 2870

Goal Programming

- Example: buying a car.
  - Start with a maximum purchase price that you do not want to exceed: this is the goal.
  - However, you will probably find a way to spend more than this amount if it is impossible to get the car you really want for your goal amount.
  - So, the goal you have in mind is not a hard constraint, that cannot be violated.
  - More accurately, it is a soft constraint, representing a target you would like to achieve.

Goal Programming

- Generally optimization techniques assume hard constraints.
  - May be too restrictive in certain situations.
- Numerous decision-making problems can be modeled accurately using goals rather than hard constraints.
  - These problems do not have one explicit objective function to be maximized or minimized over a constraint set.
  - Instead, can be stated as a collection of goals that might also include hard constraints.

Goal Programming

- Typically, goals can be achieved only at the expense of other goals.
- A hierarchy of importance needs to be established so that higher-priority goals are satisfied before lower-priority goals are addressed.
- It is not always possible to satisfy every goal so goal programming attempts to reach a satisfactory level of multiple objectives.
- The objective function tries to minimize the deviations between goals and what we can actually achieve within the given constraints.

Introduction to Goal Programming

- Goal programming (GP) is a type of mathematical programming that focuses on multiple conflicting objectives or goals.
- When applying GP methodology a target value or desired level of attainment is required for each goal.
- Either relative weights that permit combining deviations by which goals are under- or overachieved, or an absolute priority or ordinal preference ordering is required.

Introduction to Goal Programming

- The GP rationale provides a solution to the programming problem by minimizing the weighted absolute deviations (or the absolute deviations) in the specified priority order of goal attainment.
- The ordinal GP has a multi-dimensional objective function that seeks to minimize selected absolute deviations from a set of stated goals, usually within additional constraints.
- Each selected deviation in the GP objective carries an ordinal priority weight so that goals are attained (or approached as nearly as possible) in strict order of priority.

Outline

- Priorities are expressed as either ordinal or cardinal priorities.
- Ordinal priorities
  - Most important goal is satisfied first, followed by the satisfaction of as many other goals as possible, in rank order.
- Cardinal priorities
  - Assumes that goal attainment is additive and an aggregate, linear objective function is developed.

General Format

- Find $X = (x_1, x_2, \ldots, x_j, \ldots x_n)$ so as to obtain:
  min $Z = f(d^+, d^-)$
  subject to:
  (1) $AX = B + d^+ - d^-$
  (2) $CX \leq D$
  (3) $X, d^+, d^- \geq 0$

- The objective function can also be of the form:
  min $Z = \{P_1[g_1(d_1^+, d_1^-)], P_2[g_2(d_2^+, d_2^-)], \ldots, P_i[g_i(d_i^+, d_i^-)]\}$

  where:
  $g_i(d_i^+, d_i^-)$ is a linear function of deviation variables,
  P_i is the ordinal priority level associated with $g_i(d_i^+, d_i^-)$
  $i \leq m$ where m is the number of right-hand side targets.

General Format

- $X = (x_1, x_2, \ldots, x_j, \ldots x_n)$ is the solution vector.
- $f(d^+, d^-)$ is a linear, prioritized vector of the permissible deviation variables.
- d^+ is a vector of non-negative variables that are positive deviations from the objectives.
- d^- is the vector of non-negative variables that are negative deviations from the objectives.
- B is a vector of right-hand side target values, or aspiration levels, associated with the objectives.
- m is the number of target values.
- A is a matrix of activity coefficients.
- C is a matrix of resource consumption coefficients.
- D is a vector of right hand side bounds on absolute constraints.
- There are $k=m+1, m+2, \ldots, M$ absolute constraints.

General Format

- The priority levels, P_i, have a strict interpretation as the preemptive priority associated with the ith objective.
  - Thus, P_1, is the top objective.
- Furthermore, $P_i \gg P_{i+1} \gg P_{i+2} \ldots$
  - This means that satisfying priority level i objectives is preferred to satisfying any other lower priority objectives.
  - The solution procedure does not find a global optimum satisfying all constraints, as in LP.
  - In GP, find a feasible set of the best solutions to the priority 1 level sub-problem.
  - Then within these solutions GP finds that subset of solutions that best solves the priority level 2 sub-problem, etc.

General Format

- GP permits intentional positive, negative, or zero valued deviations away from the numerically valued goals.
- However, both d_i^+, d_i^- cannot simultaneously take on positive values in the simplex procedure.
- Hence, only d_i^+ or d_i^- can be in solution in the same problem tableau.

Example

- Schedule the production of two products on three machines.
  - x =product 1
  - y = product 2

Product	Machines			
	1	2	3	
1	0.2 hr	0.4 hr	0.3 hr	Time/unit
2	0.4 hr	0.1 hr	0.3 hr	Time/unit
Capacity	40 hr	40 hr	40 hr	Time/week
Product	Maximum demand			Profit
1	80/week			$5/unit
2	80/week			$20/unit

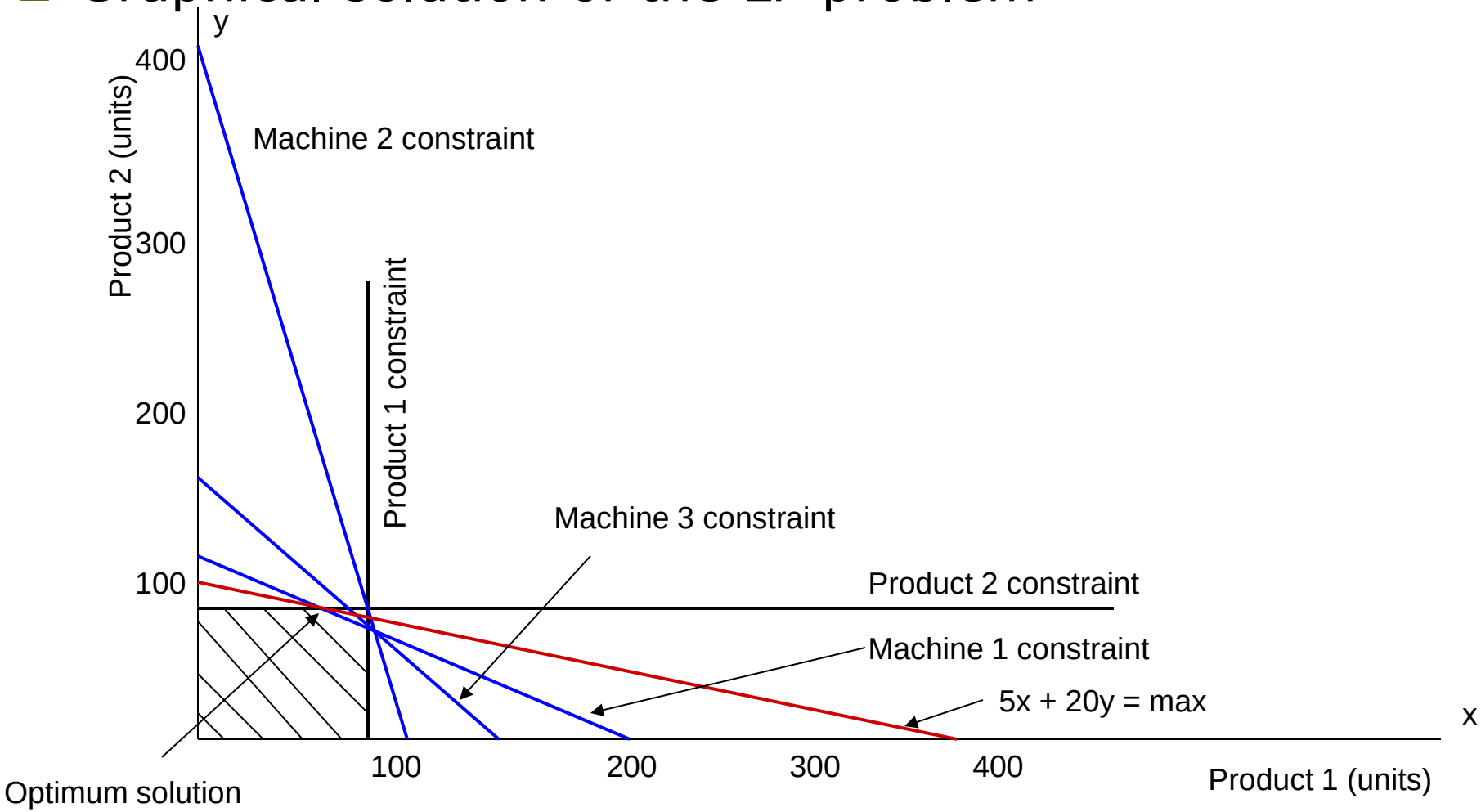
Example

- LP formulation of the production scheduling problem.

$$
\begin{aligned}
&\text{Max} && 5x + 20y \\
&\text{Subject to} && \\
& && 0.2x + 0.4y \leq 40 \\
& && 0.4x + 0.1y \leq 40 \\
& && 0.3x + 0.3y \leq 40 \\
& && x \leq 80 \\
& && y \leq 80 \\
& && x \geq 0 \\
& && y \geq 0
\end{aligned}
$$

Example

- Graphical solution of the LP problem



Example

- Optimum solution
  - x = 40
  - y = 80
- Unused machine capacity
  - Machine 1 = 0 hours/week
  - Machine 2 = 16 hours/week
  - Machine 3 = 4 hours/week
- Comment
  - Machine 2 is only 60% utilized

Example

- Manager is concern not only with the utilization of the machines but also with the profits realized.
- Manager recognizes also that the market forecasts of 80 units per week could be in error.
- Manager would prefer to have the three machines equally loaded, but does not want to sacrifice profits.
- Maximum profit condition ($x = 40$, $y = 80$) results in a profit of $1,800.
- Manger feels that a satisfactory minimum profit would be $1,500.

Example

- GP formulation of the scheduling problem

Min $(d_1^-, d_2^- + d_3^- + d_4^-)$

Subject to

$$5x + 20y + d_1^- - d_1^+ = \$1{,}500$$
$$0.2x + 0.4y + d_2^- - d_2^+ = 40$$
$$0.4x + 0.1y + d_3^- - d_3^+ = 40$$
$$0.3x + 0.3y + d_4^- - d_4^+ = 40$$
$$x \leq 80$$
$$y \leq 80$$
$$x, y, d_j^-, d_j^+ \geq 0$$

Example

- Slack variable d_j^- and surplus variable d_j^+ measure the deviation from the right-hand side of the constraints.
- Since the upper limit of 40 hours of production capacity cannot be exceeded, the surplus variables can be omitted for j = 2, 3, and 4.
- The objective function is expressed using an ordinal scaling of the profit goal (make at least $1,500 profit) and the machine balance goal (minimize the sum of unused capacities).

Example

- Resulting GP formulation

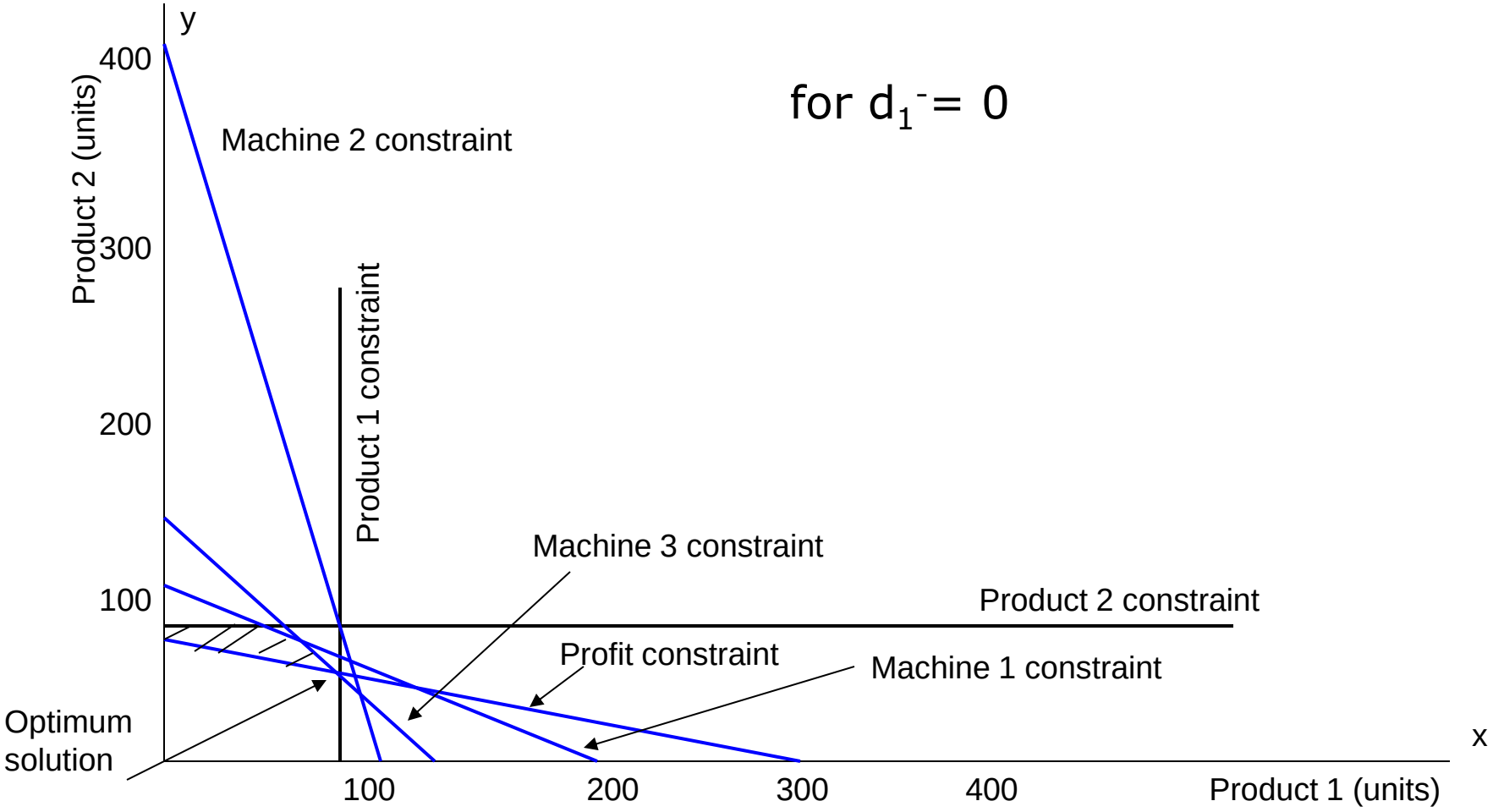
$$\text{Min} \quad (d_1^-, d_2^- + d_3^- + d_4^-)$$

Subject to

$$5x + 20y + d_1^- - d_1^+ = \$1{,}500$$
$$0.2x + 0.4y + d_2^- = 40$$
$$0.4x + 0.1y + d_3^- = 40$$
$$0.3x + 0.3y + d_4^- = 40$$
$$x \leq 80$$
$$y \leq 80$$
$$x, y, d_j^-, d_j^+ \geq 0$$

Example

- The set of optimum solutions is determined using the following objective: minimize d_1^-.



for $d_1^- = 0$

Example

- Solve the linear programming problem

$$
\begin{aligned}
\text{Minimize} \quad & d_2^- + d_3^- + d_4^- \\
\text{Subject to} \quad & 0.2x + 0.4y + d_2^- = 40 \\
& 0.4x + 0.1y + d_3^- = 40 \\
& 0.3x + 0.3y + d_4^- = 40 \\
& 5x + 20y - d_1^+ = \$1{,}500 \\
& x \le 80 \\
& y \le 80 \\
& x, y, d_j^-, d_j^+ \ge 0
\end{aligned}
$$

Example

- Solution
  - $x = 700/9$
  - $y = 500/9$
  - $d_2^- = 20/9$
  - $d_3^- = 10/3$
  - $d_4^- = 0$
  - $d_1^+ = 0$
  - Profit = $1,500
  - Machine utilizations: 94.44%, 91.67%, and 100%.

Example

- An alternative expression of the machine balance goal is to minimize the maximum difference in the d_j^- values.

Min $\max(|d_2^- - d_3^-|, |d_2^- - d_4^-|, |d_3^- - d_4^-|)$

Subject to

$$0.2x + 0.4y + d_2^- = 40$$
$$0.4x + 0.1y + d_3^- = 40$$
$$0.3x + 0.3y + d_4^- = 40$$
$$5x + 20y - d_1^+ = \$1{,}500$$
$$x \le 80$$
$$y \le 80$$
$$x, y, d_j^-, d_j^+ \ge 0$$

Example

- Let z be the maximum absolute deviation among the slack variables.

Min z

Subject to

$$z \geq |d_2^- - d_3^-|$$
$$z \geq |d_2^- - d_4^-|$$
$$z \geq |d_3^- - d_4^-|$$
$$0.2x + 0.4y + d_2^- = 40$$
$$0.4x + 0.1y + d_3^- = 40$$
$$0.3x + 0.3y + d_4^- = 40$$
$$5x + 20y - d_1^+ = \$1{,}500$$
$$x \leq 80$$
$$y \leq 80$$
$$x, y, d_j^-, d_j^+ \geq 0$$

Example

- First 3 constraints ensure that the minimum value of z will equal the maximum absolute deviation among the slack variables.
- A constraint of the form

$$|u - v| \leq w$$

can be written as two linear constraints

$$u - v \leq w$$

$$u - v \geq w$$

Example

- Optimization problem can be written as

Min z

Subject to

$$
\begin{aligned}
d_2^- - d_3^- - z &\le 0 \\
-d_2^- + d_3^- - z &\le 0 \\
d_2^- - d_4^- - z &\le 0 \\
-d_2^- + d_4^- - z &\le 0 \\
d_3^- - d_4^- - z &\le 0 \\
-d_3^- + d_4^- - z &\le 0 \\
0.2x + 0.4y + d_2^- &= 40 \\
0.4x + 0.1y + d_3^- &= 40 \\
0.3x + 0.3y + d_4^- &= 40 \\
5x + 20y - d_1^+ &= \$1{,}500 \\
x &\le 80 \\
y &\le 80 \\
x, y, d_j^-, d_j^+ &\ge 0
\end{aligned}
$$

Example

- Solution
  - $x = 700/9$
  - $y = 500/9$
  - $d_1^+ = 0$
  - $d_2^- = 20/9$
  - $d_3^- = 10/3$
  - $d_4^- = 0$
  - $z = 10/3$
  - Profit = $1,500
  - Machine utilizations: 94.44%, 91.67%, and 100%.
- Same solution for both
  - Minimax objective
  - Objective of minimizing the sum of the deviations.